

Chapter 7

Aqueous Solutions and Chemical Equilibria

Fundamentals of Analytical Chemistry, Skoog, et al., 10th Ed., 2022

Chapter 7 Problems:

7-8, 7-10, 7-15

The Chemical Composition of Aqueous Solutions

Classifying Solutions of Electrolytes

Strong Electrolytes: essentially ionize completely

Strong Acids, alkali and alkaline-earth hydroxides, most salts

A salt is produced in the reaction of an acid and base

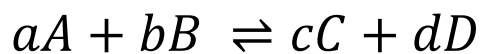
Weak Electrolytes: partially ionize

Many inorganic Acids (carbonic), most organic acids, ammonia and most organic bases

Chemical Equilibrium

Equilibrium-Constant Expression

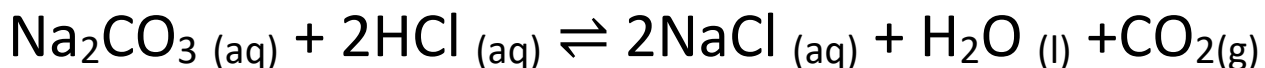
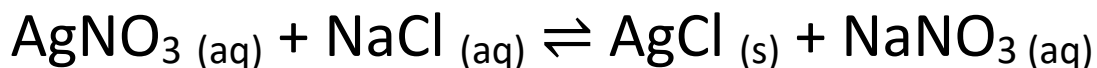
Algebraic equation that describes the concentration relationships among reactants and products at equilibrium



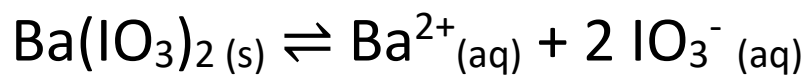
$$K = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

Terms are:

1. Molar concentration if they represent dissolved solutes
2. Partial pressures in atmospheres if they are gas-phase reactants or products
3. If a reactant or product is a pure liquid, a pure solid, or the solvent present in excess, no term for this species appears in the equilibrium constant expression.



Using Solubility-Product Constants



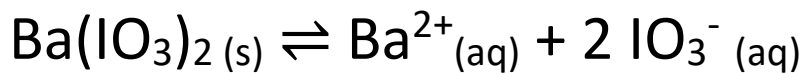
$$k_{sp} = [\text{Ba}^{2+}] [\text{IO}_3^-]^2$$

You can find the k_{sp} values in appendix 2

The solubility of a Precipitate in Pure Water

1. What mass (in grams) of $\text{Ba}(\text{IO}_3)_2$ (MM = 487 g/mol) can be dissolved in 500 mL of water at 25°C?

K_{sp} for $\text{Ba}(\text{IO}_3)_2$ is 1.57×10^{-9} (see appendix 2).



$$k_{sp} = [\text{Ba}^{2+}] [\text{IO}_3^-]^2 = 1.57 \times 10^{-9}$$

The equation describing the equilibrium reveals that 1 mole of Ba^{2+} is formed for each mole of $\text{Ba}(\text{IO}_3)_2$ that dissolves.

The Effect of a Common Ion on the Solubility of a Precipitate

Example 9-4

2. Calculate the molar solubility of $\text{Ba}(\text{IO}_3)_2$ in a solution that is 0.0200 M $\text{Ba}(\text{NO}_3)_2$.

Example 7-5

3. Calculate the molar solubility of $\text{Ba}(\text{IO}_3)_2$ in a solution prepared by mixing 200 mL of 0.0100 $\text{Ba}(\text{NO}_3)_2$ with 100 mL of 0.100 M NaIO_3 .

First establish whether either reactant is present in excess at equilibrium. The amounts taken are

$$\# \text{ mmol Ba}^{2+} = 200 \text{ mL} \times 0.0100 \text{ mmol/mL} = 2.00 \text{ mmol Ba}^{2+}$$

$$\# \text{ mmol IO}_3^- = 100 \text{ mL} \times 0.100 \text{ mmol/mL} = 10.0 \text{ mmol IO}_3^-$$

If the formation of $\text{Ba}(\text{IO}_3)_2$ is complete:

$$\# \text{ mmol excess NaIO}_3 = 10.0 \text{ mmol IO}_3^- - [2 \text{ mmol Ba}^{2+} \times (2.00 \text{ IO}_3^- / 1 \text{ mmol Ba}^{2+})] = 6.0 \text{ mmol IO}_3^-$$